

JPA

- JPA is **only a specification**.
- It defines rules for object-relational mapping (ORM).
- It cannot work by itself.
- It requires an implementation such as Hibernate.

Hibernate

- Hibernate is an **ORM framework**.
- It implements JPA.
- Converts Java objects into database tables automatically.
- Provides additional features like caching, lazy loading, and HQL.

Spring Data JPA

- Spring Data JPA is built on top of JPA.
- Reduces database code significantly.
- Uses repositories to perform CRUD operations.

➔ **JPA is a specification.**

➔ **Hibernate** is the most popular **implementation of JPA**.

➔ **Spring Data JPA** uses **JPA** and makes database operations much easier by reducing boilerplate code.